

第十三讲：基于CNN的目标检测

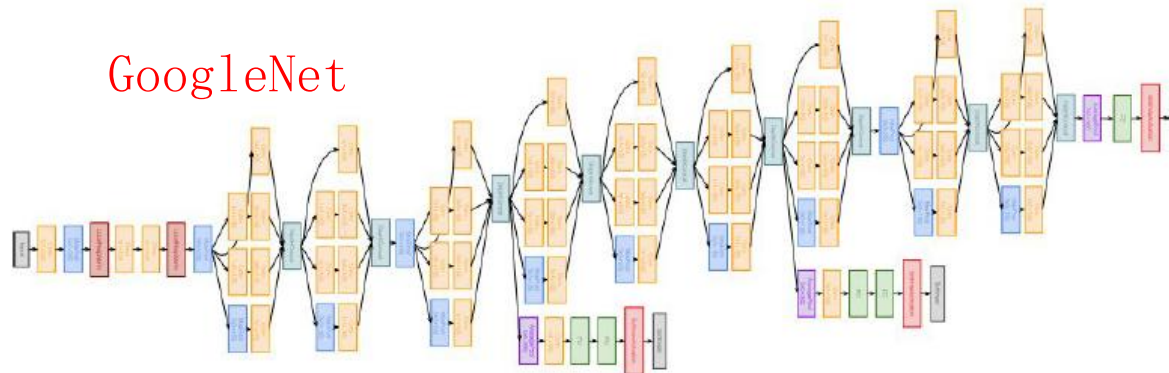


授课人：王国泰 wgt_cv@163.com

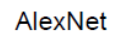
电子科技大学机械与电气工程学院

2026-4-14



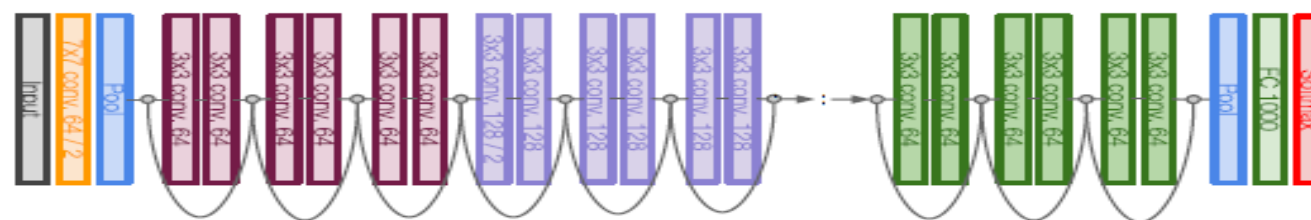


ResNet



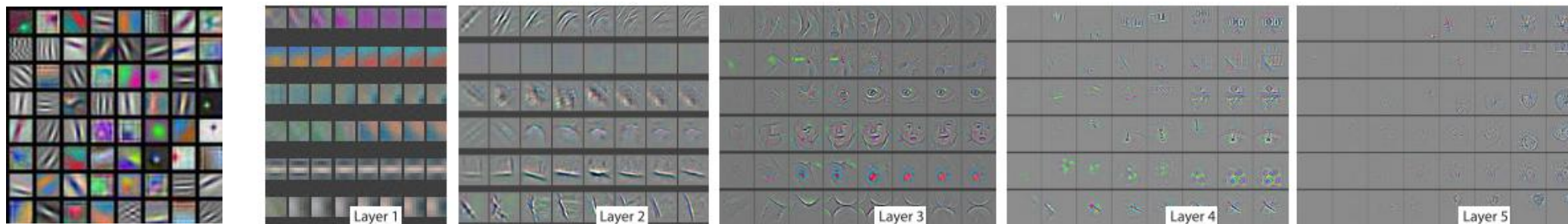
VGG16

VGG19

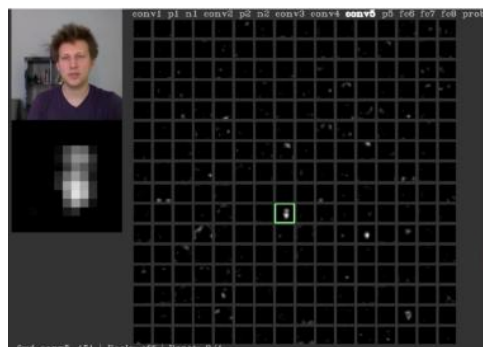


知识回顾：对CNN的理解

卷积核的可视化



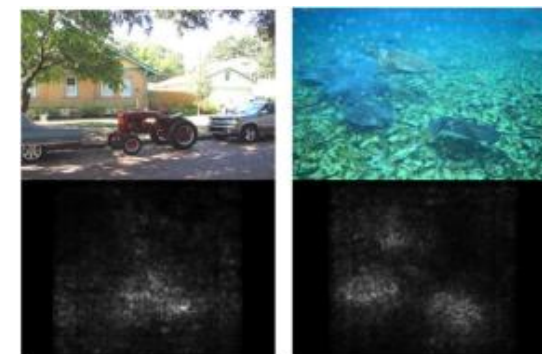
特征图的可视化



某一层响应最大的图像



图像梯度分析



图像识别任务：从图像级别到像素级别



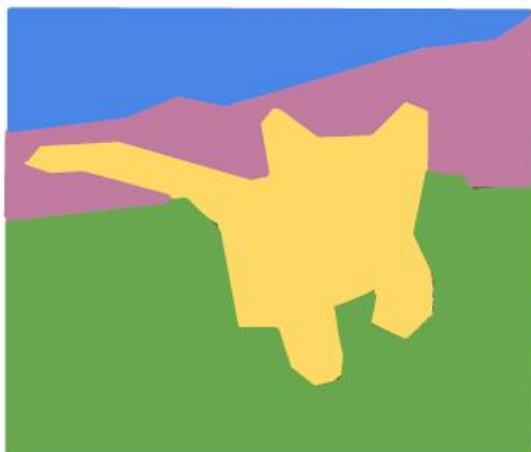
Classification
分类



CAT

No spatial extent

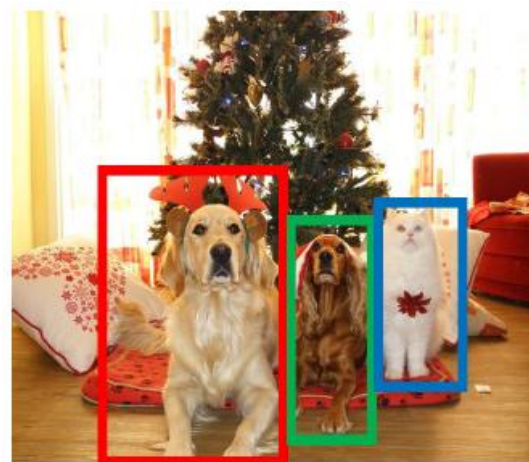
Semantic segmentation
场景分割



CAT, GRASS,
TREE, SKY

No objects, just pixels

Object detection
物体检测



DOG, DOG, CAT

Multiple Object

Instance segmentation
实例分割



DOG, DOG, CAT

目录

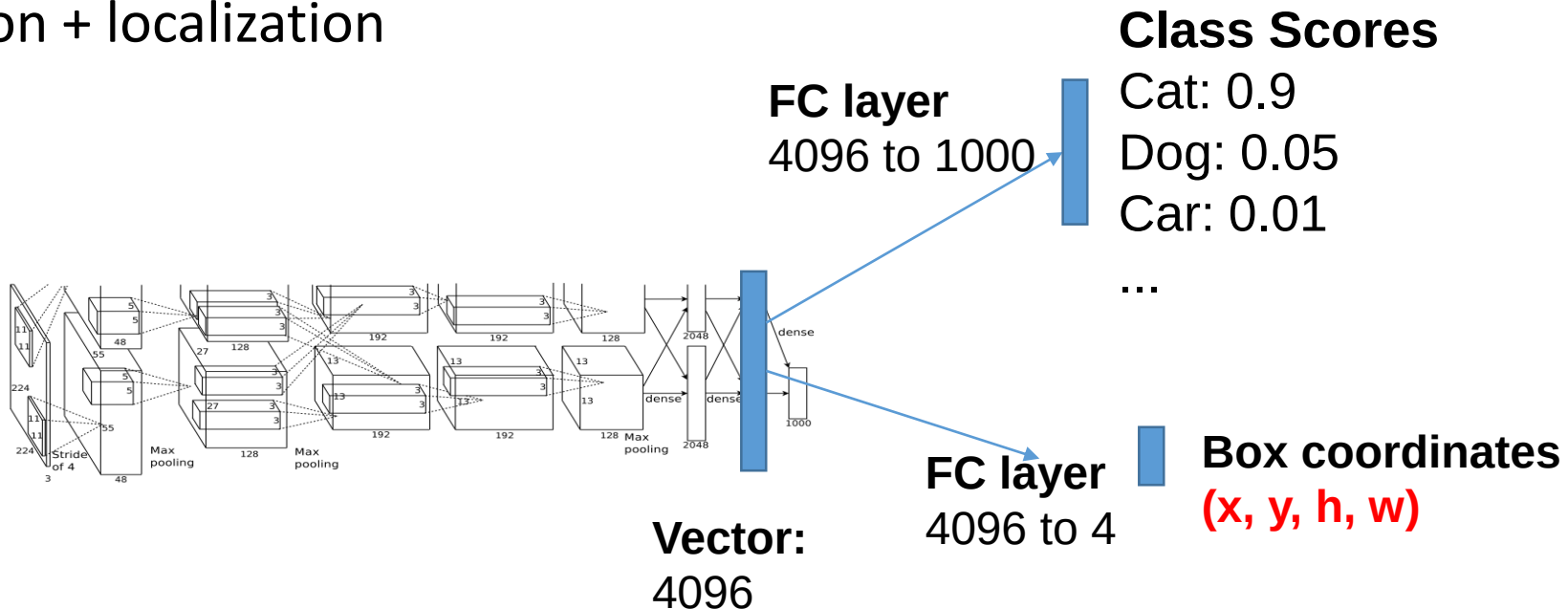
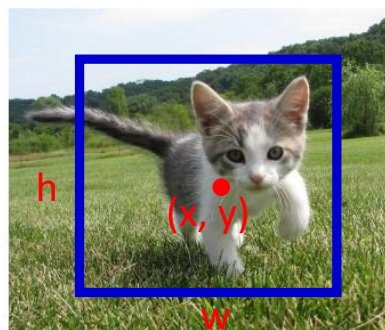


- 目标检测
 - R-CNN
 - Fast R-CNN
 - Faster R-CNN
 - SSD, YOLO
- 图像分割
 - FCN
 - UNet

Object Detection 目标检测

Single object detection 单个目标的检测

- Classification + localization



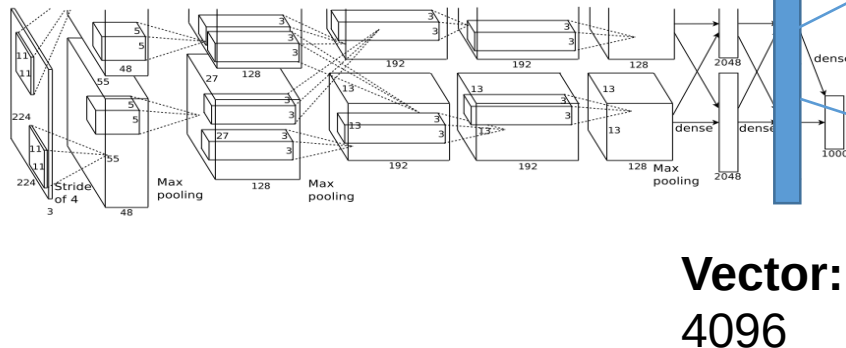
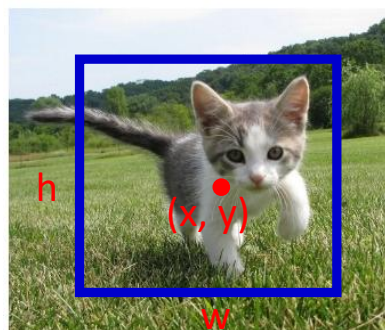
将定位问题视作回归问题

Object Detection 目标检测



Single object detection 单个目标的检测

- Classification + localization



FC layer
4096 to 1000

Class Scores

Cat: 0.9
Dog: 0.05
Car: 0.01
...

Correct label
Cat

Softmax Loss

分类损失

边框回归损失

FC layer
4096 to 4

Box coordinates
 (x, y, h, w)

L2 loss

Real coordinates
 (x', y', h', w')

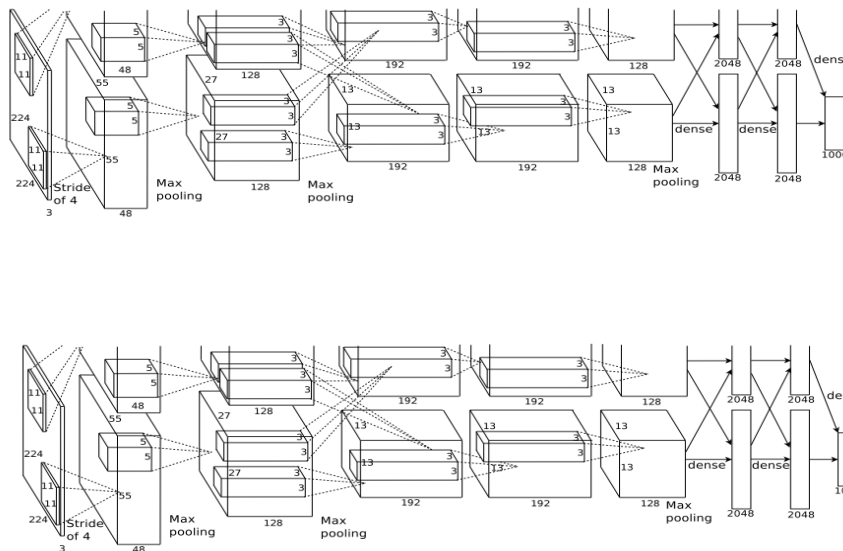
将定位问题视作回归问题

总损失函数 = 分类损失 + 边框回归损失

Object Detection 目标检测

Multi-object detection 多个目标的检测

- Classification + localization

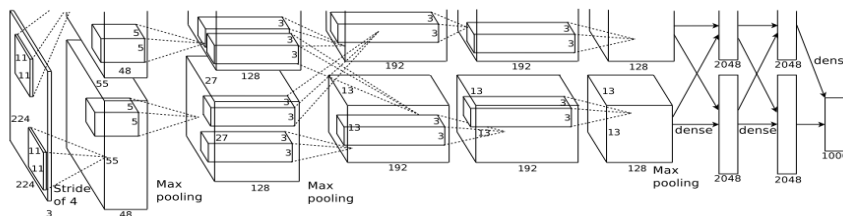


DOG: (x, y, w, h)

DOG: (x, y, w, h)

CAT: (x, y, w, h)

12个预测值



DUCK: (x, y, w, h)

DUCK: (x, y, w, h)

....

42个预测值

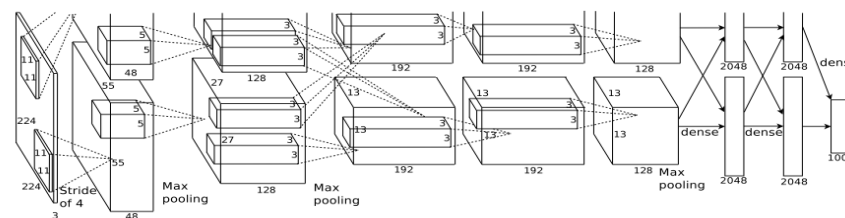
每个图像中包含的物体个数不一样！

无法通过预设个数的节点进行预测。

Object Detection 目标检测

Basic idea: sliding window classification 滑动窗口分类

- Apply a CNN to many different crops of the Image
- CNN classifies each crop as object or background



Dog? NO

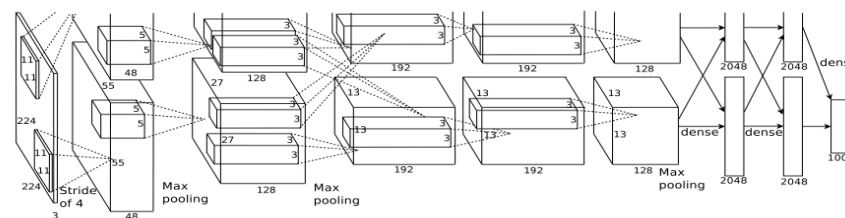
Cat? NO

Background? **YES**

Object Detection 目标检测

Basic idea: sliding window classification 滑动窗口分类

- Apply a CNN to many different crops of the Image
- CNN classifies each crop as object or background



Dog? **Yes**

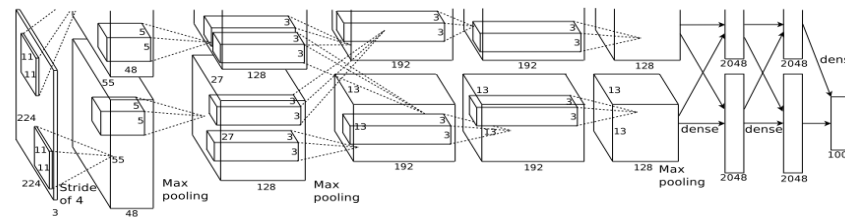
Cat? NO

Background? No

Object Detection 目标检测

Basic idea: sliding window classification 滑动窗口分类

- Apply a CNN to many different crops of the Image
- CNN classifies each crop as object or background



Dog? No

Cat? **Yes**

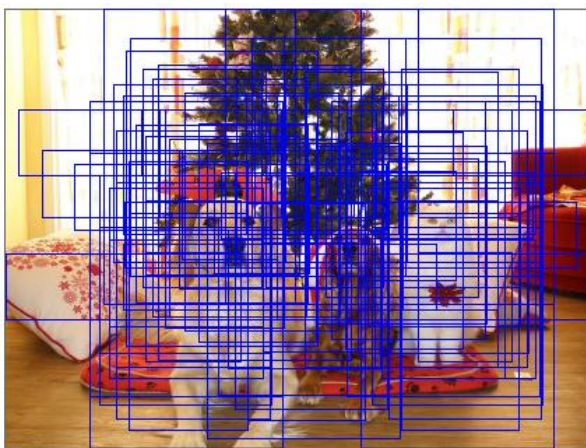
Background? No

Q: what is the problem of this approach?

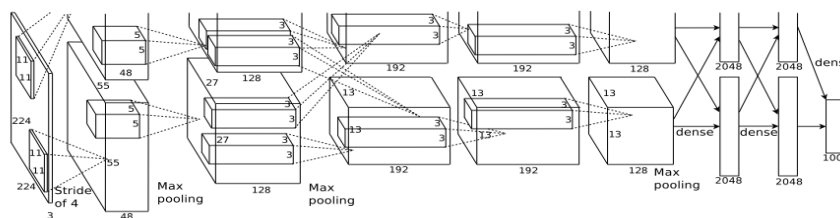
Object Detection 目标检测

1, Basic idea: sliding window classification 滑动窗口分类

- Apply a CNN to many different crops of the Image
- CNN classifies each crop as object or background



Problem: Need to apply CNN to huge number of locations, scales, and aspect ratios, **very computationally expensive!**



Dog? No

Cat? **Yes**

Background? No

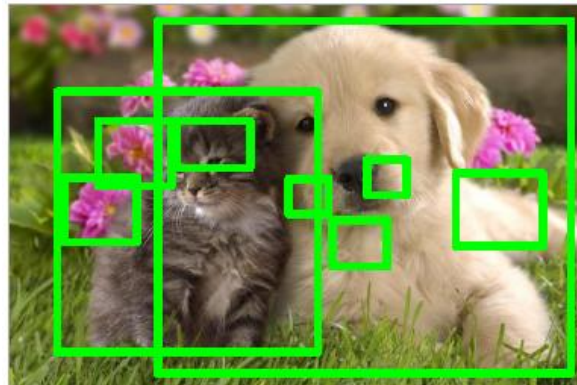
需要考虑不同的位置、尺度、长宽比，计算量大！

Object Detection 目标检测



Region proposal 区域提议

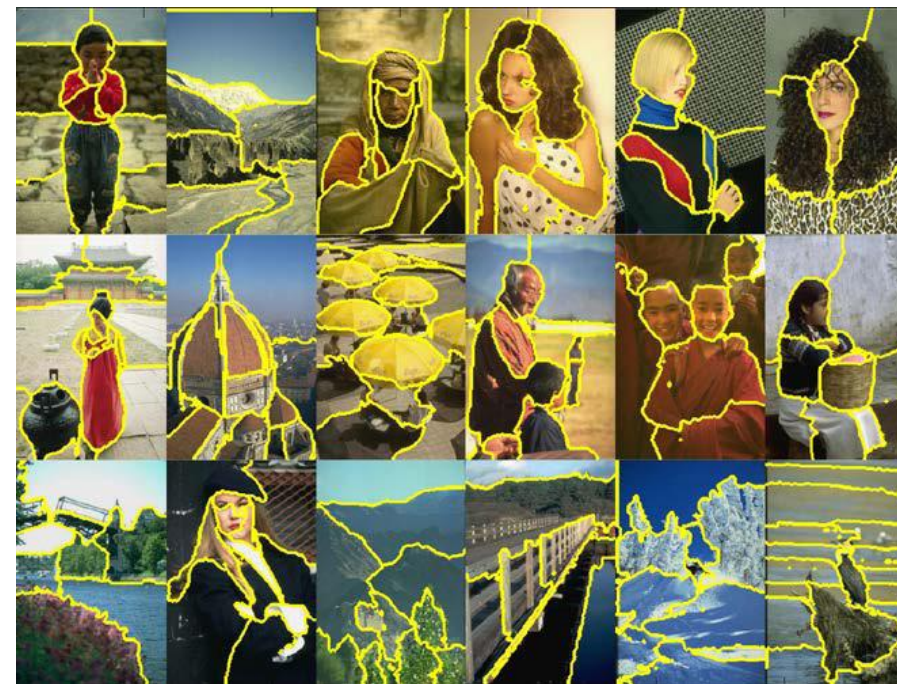
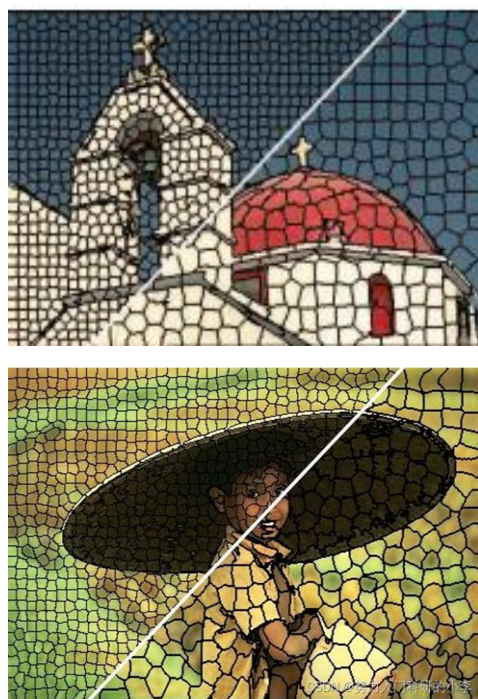
- Find “blobby” image regions that are likely to contain objects
- Fast to run; e.g. Selective Search gives 2000 region proposals in a few seconds on CPU



Object Detection 目标检测



Region proposal by Clustering 通过聚类产生区域提议



SLIC Super-pixel 超像素

Object Detection 目标检测



Region proposal: Selective Search by Hierarchical Grouping



通过分层次聚类，产生不同尺度的区域提议

Algorithm 1: Hierarchical Grouping Algorithm

Input: (colour) image

Output: Set of object location hypotheses L

Obtain initial regions $R = \{r_1, \dots, r_n\}$ using Felzenszwalb and Huttenlocher (2004) Initialise similarity set $S = \emptyset$;

foreach Neighbouring region pair (r_i, r_j) **do**

 Calculate similarity $s(r_i, r_j)$;

$S = S \cup s(r_i, r_j)$;

while $S \neq \emptyset$ **do**

 Get highest similarity $s(r_i, r_j) = \max(S)$;

 Merge corresponding regions $r_t = r_i \cup r_j$;

 Remove similarities regarding r_i : $S = S \setminus s(r_i, r_*)$;

 Remove similarities regarding r_j : $S = S \setminus s(r_*, r_j)$;

 Calculate similarity set S_t between r_t and its neighbours;

$S = S \cup S_t$;

$R = R \cup r_t$;

Extract object location boxes L from all regions in R ;

Object Detection

R-CNN

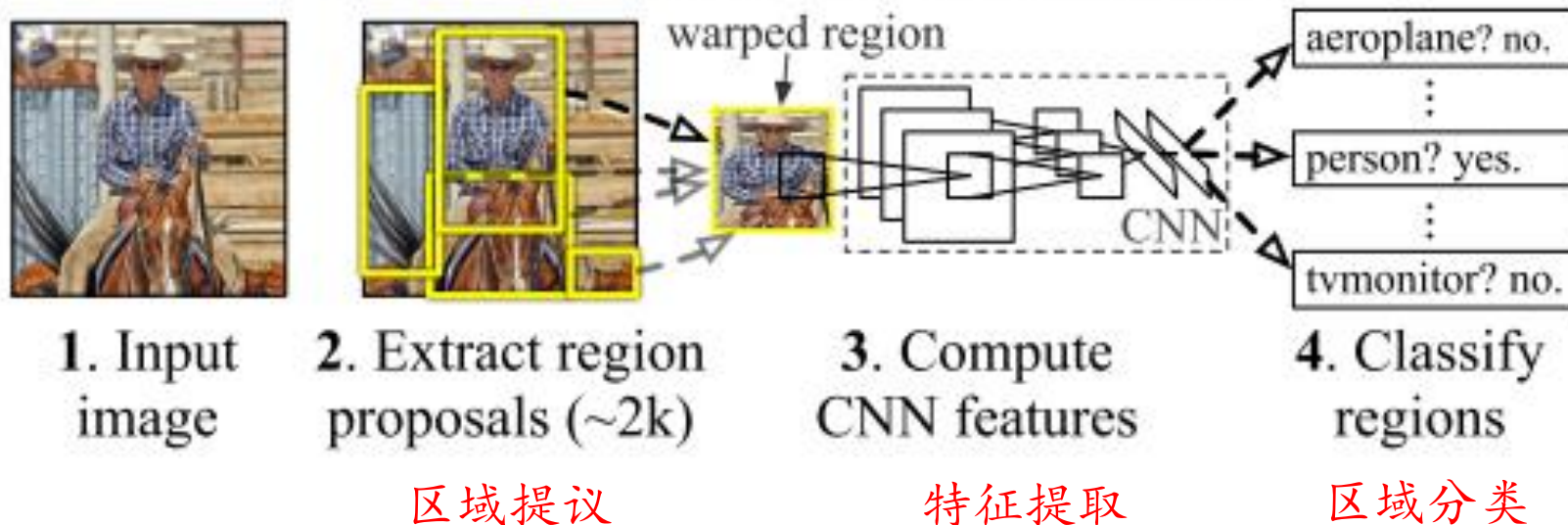
Rich feature hierarchies for accurate object detection and semantic segmentation

[R Girshick, J Donahue, T Darrell... - ... and pattern recognition, 2014 - openaccess.thecvf.com](#)

... **Object detection** with R-CNN Our **object detection** system consists of three modules. The first generates category-independent region proposals. These proposals define the set of ...

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R-CNN: *Regions with CNN features*

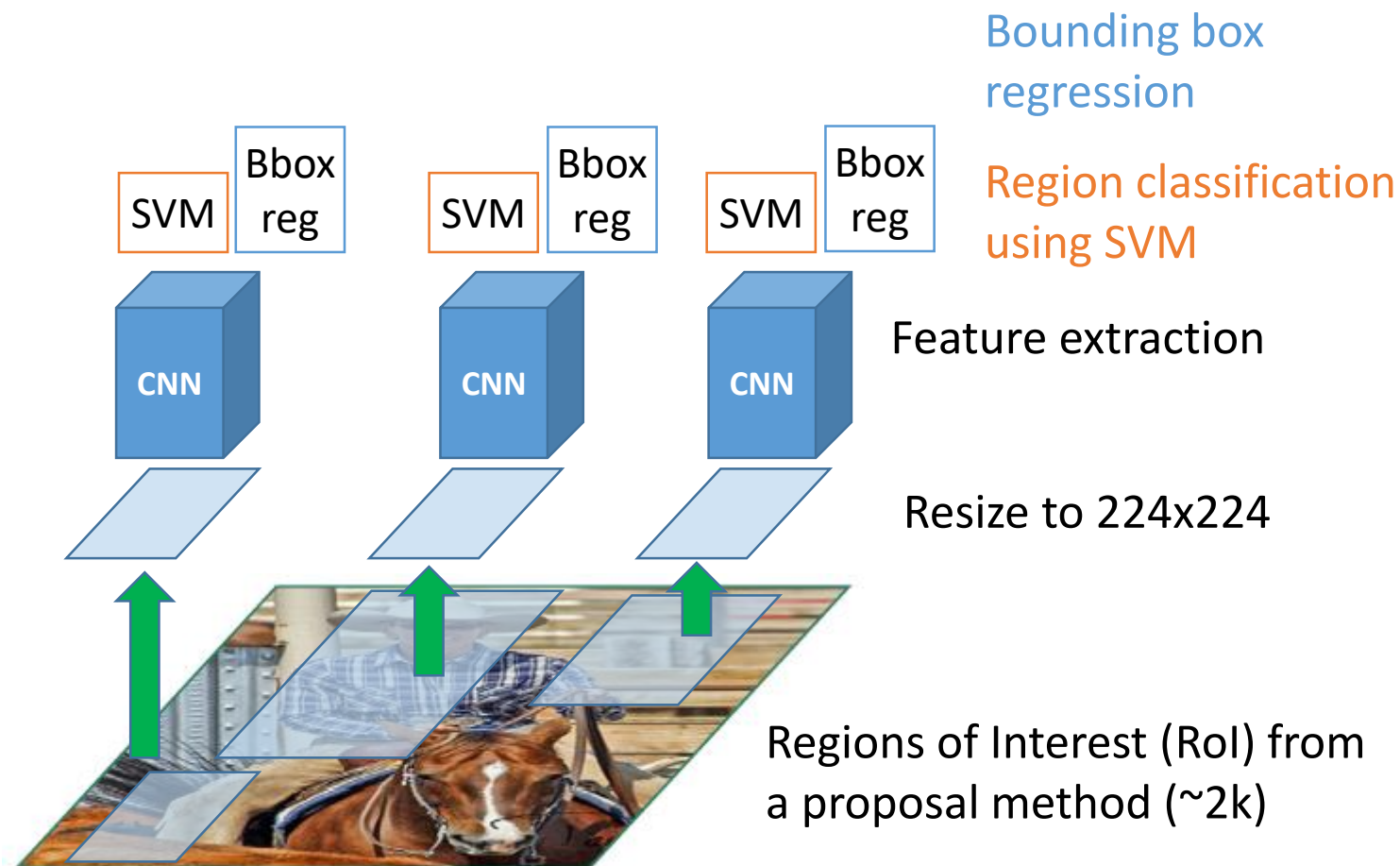


Object Detection

R-CNN

Problem: Very slow! Need to do ~2k independent forward passes for each image!

对大量重叠区域进行特征提取时
存在重复的计算



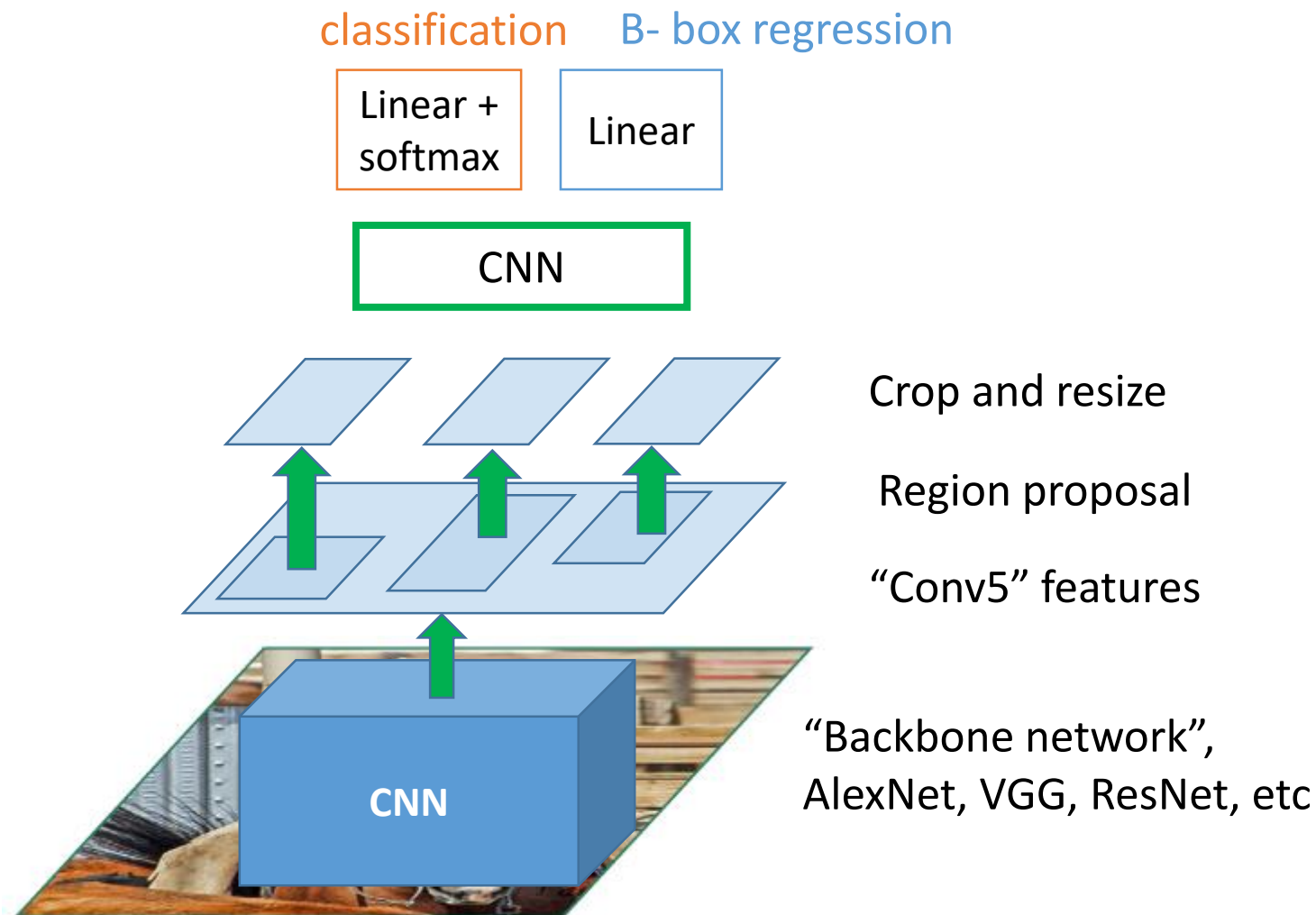
Object Detection

Fast R-CNN

Idea: Pass the image through convnet before cropping! Crop the conv feature instead!

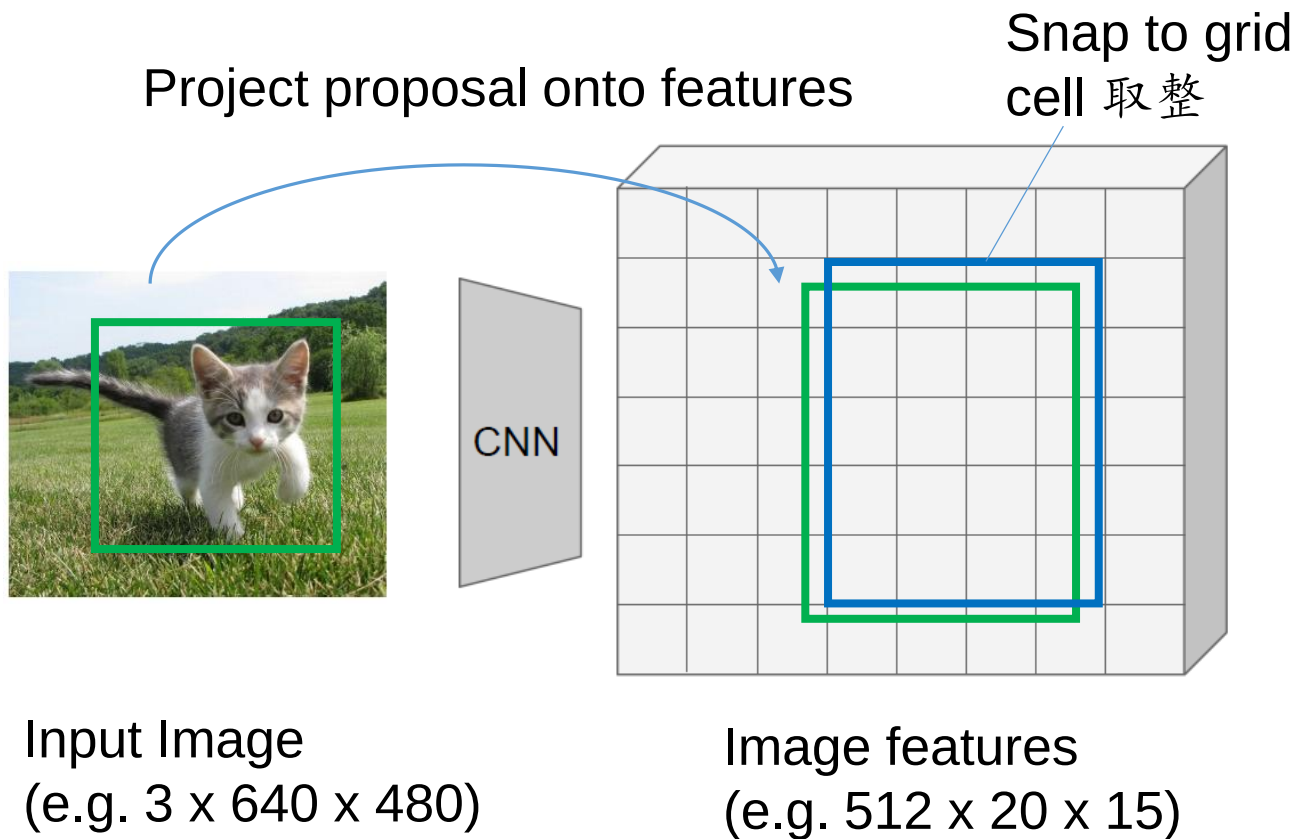
思路：先将整个图像输入CNN进行特征提取，然后再从相应的区域中提取特征

使用CNN进行分类和回归



Object Detection

Fast R-CNN with ROI pool

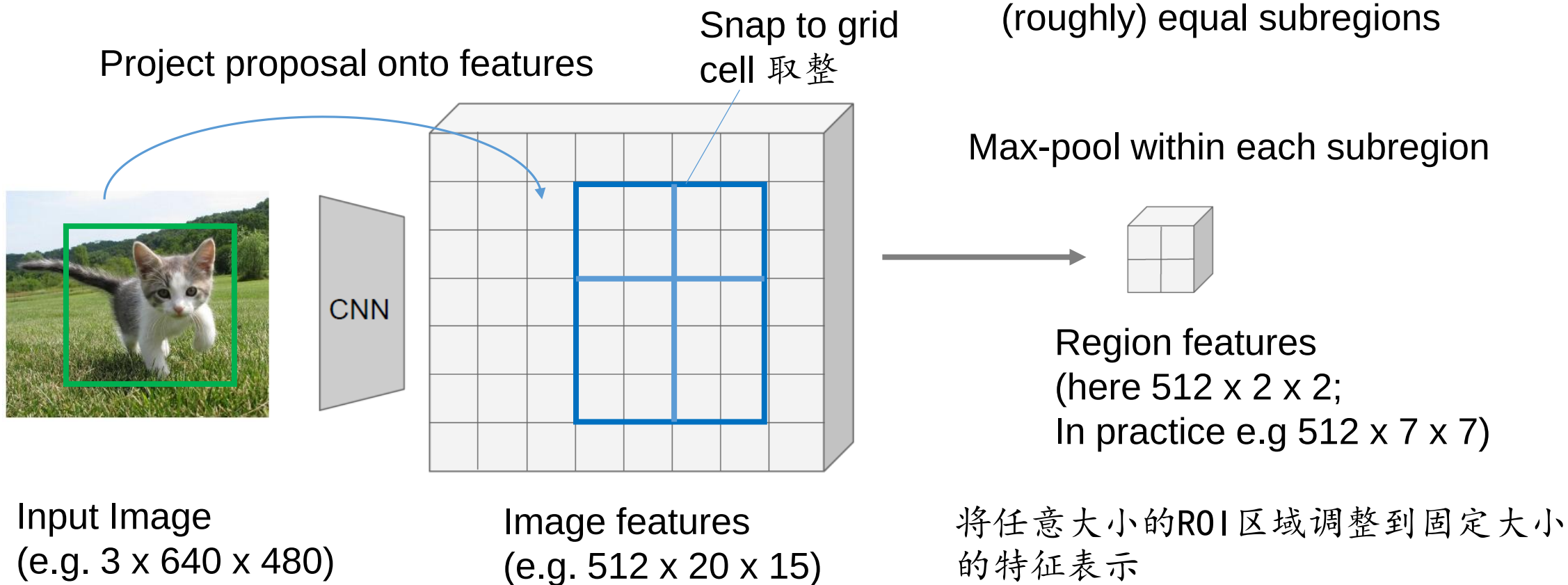


Q: how do we resize the 512 x 20 x 15 region to, e.g., a 512 x 2 x 2 tensor?

如何将任意大小的ROI区域调整到固定大小?

Object Detection

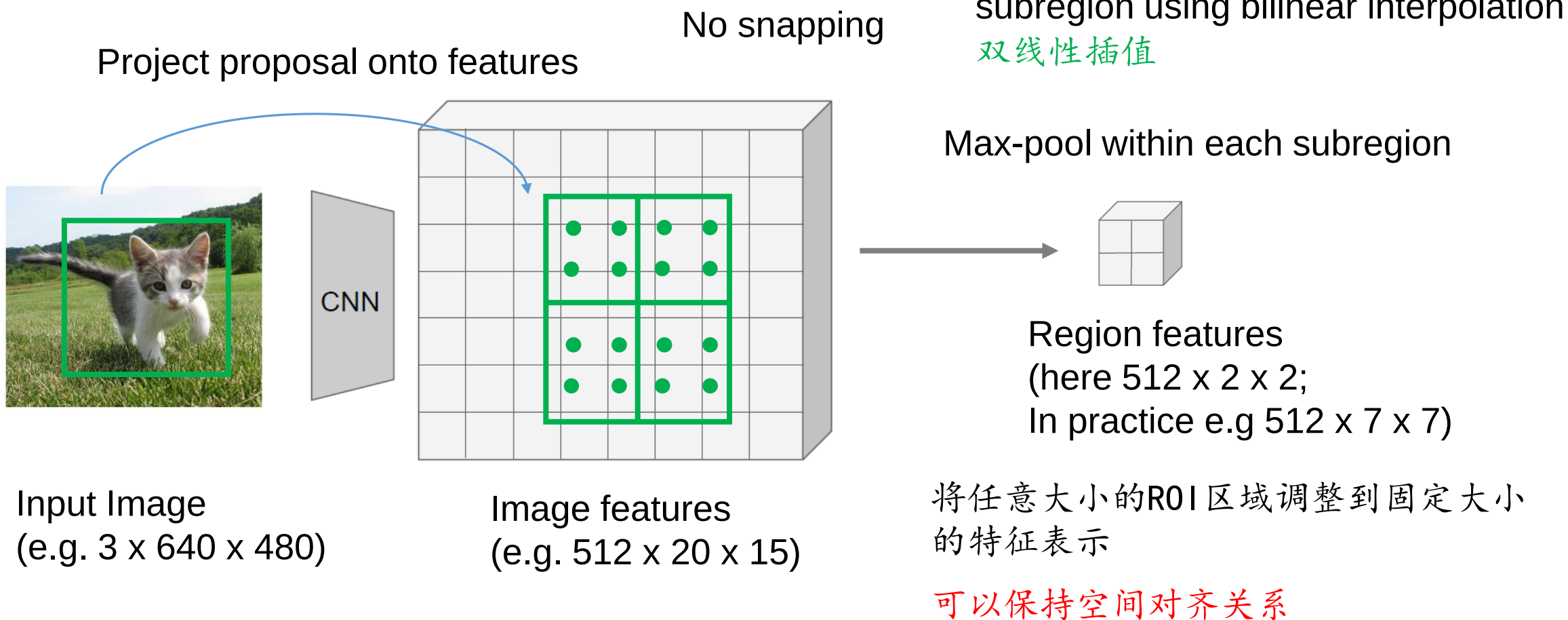
Fast R-CNN with ROI pool



Problem: Region features slightly misaligned

Object Detection

ROI pool vs ROI Align



Object Detection

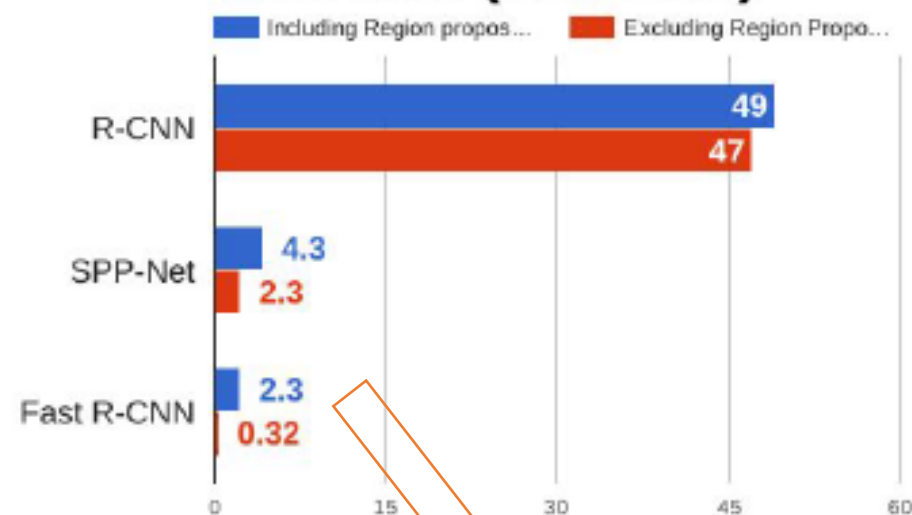


R-CNN vs Fast R-CNN

Training time (Hours)



Test time (seconds)



获取区域提议的时间占了绝大部分
(在CPU上计算)

Object Detection

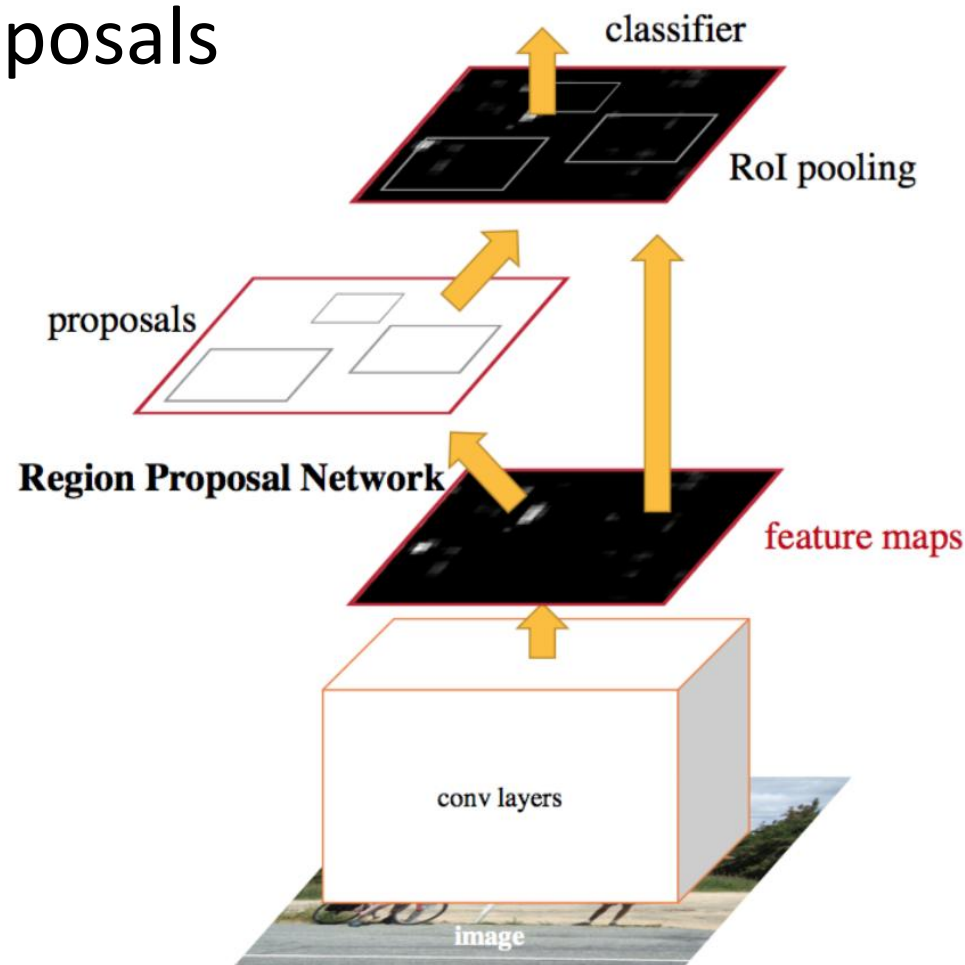
Faster R-CNN: Make CNN do proposals

直接使用CNN来预测区域提议

Region Proposal Network (RPN) to predict proposals from features

Otherwise same as Fast R-CNN:

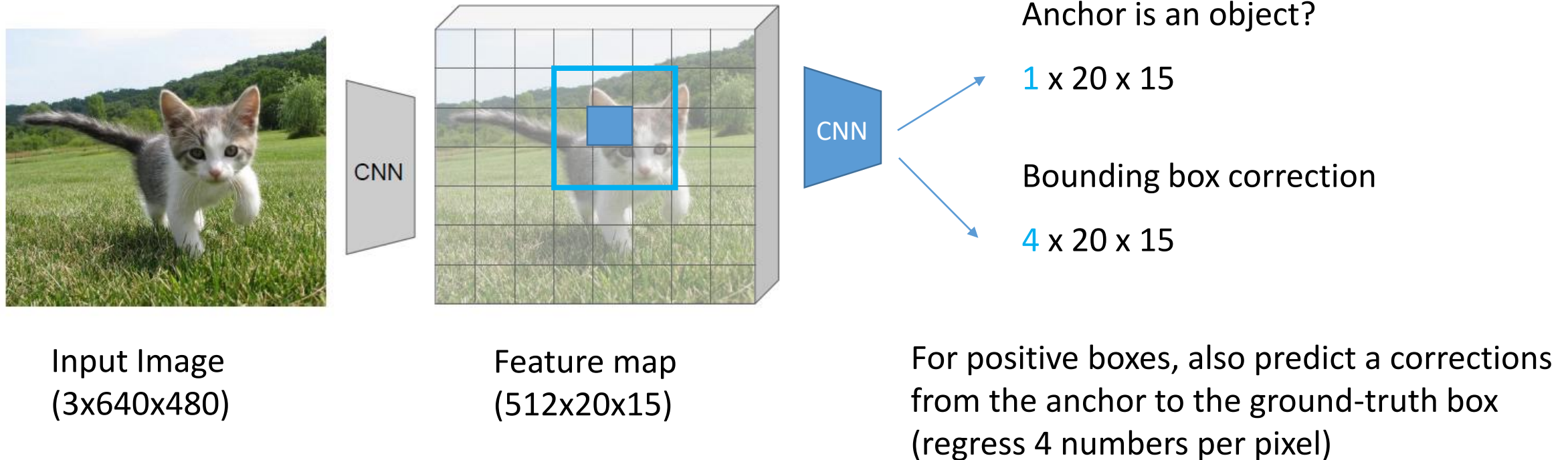
- Crop features for each proposal
- classify each one



Object Detection

Region proposal network

Imagine an **anchor box** of fixed size at each point in the feature map

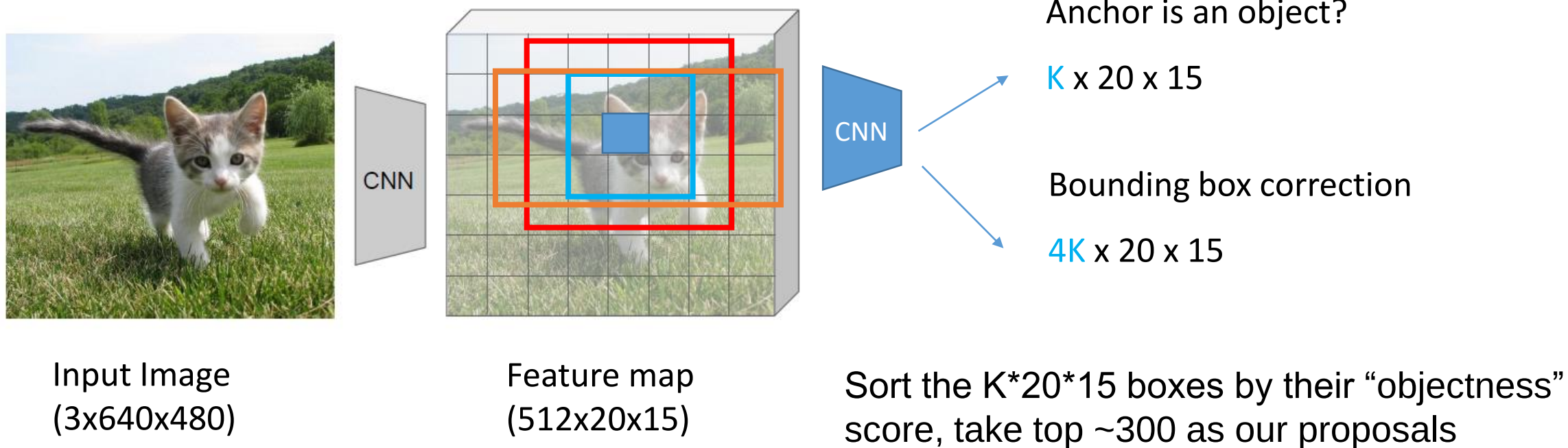


Object Detection

Region proposal network

对于每个像素，考虑K个以其为中心的预设框

use K different anchor boxes of different size / scale at each point



Object Detection

3, Faster R-CNN: Make CNN do proposals

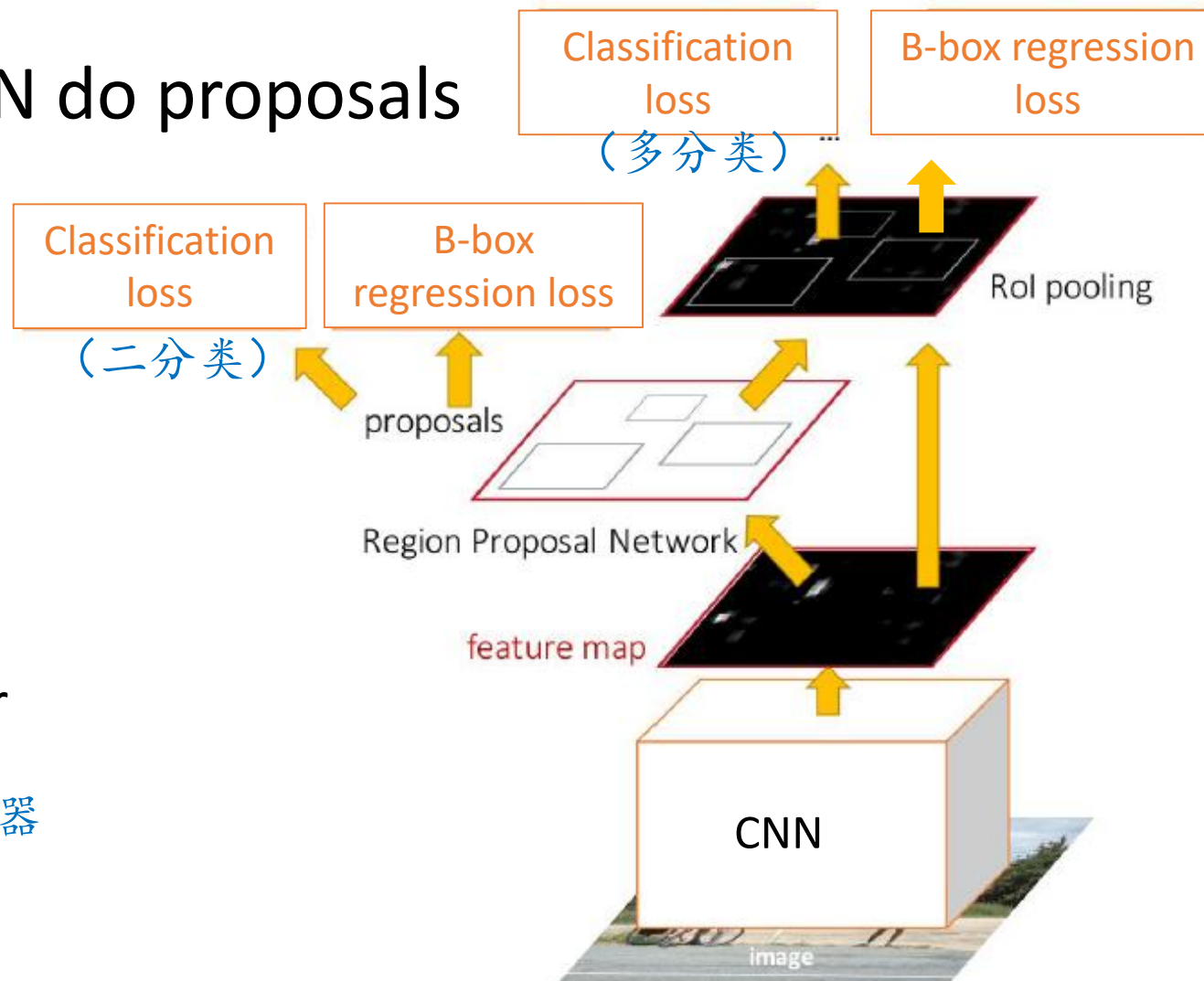
Jointly train 4 losses:

1. RPN classify object / not object
2. RPN regress box coordinates
3. Final classification score (object classes)
4. Final box coordinates

Faster R-CNN is a two-stage detector

- Obtain region proposal
- Predict each region

两阶段检测器



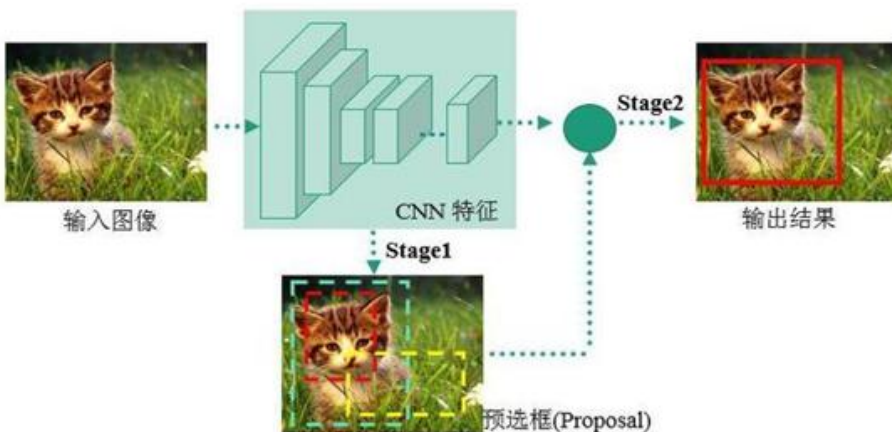
Object Detection

从两阶段检测到单阶段检测



One Stage结构

速度快，适合实时检测



Two Stage结构

精度高，但速度慢

Object Detection

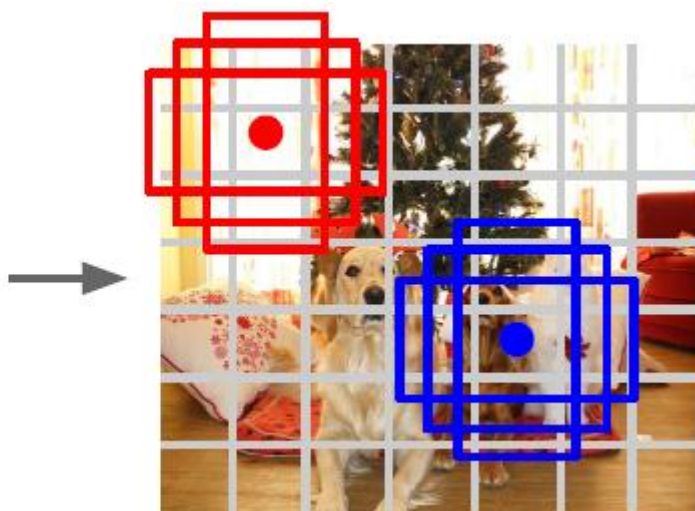
Single-stage detectors: YOLO/SSD/RetinaNet

单阶段检测器



Input Image
(3xHxW)

通过预设锚点+每个锚点
处若干个预设边界框



Divide image into grid (7x7)

Assume each cell has
B=3 base boxes

Within each grid cell:

- Regress from each of B base boxes with 5 numbers:
(dx, dy, dh, dw, **confidence**)
- Predict score for each of C classes

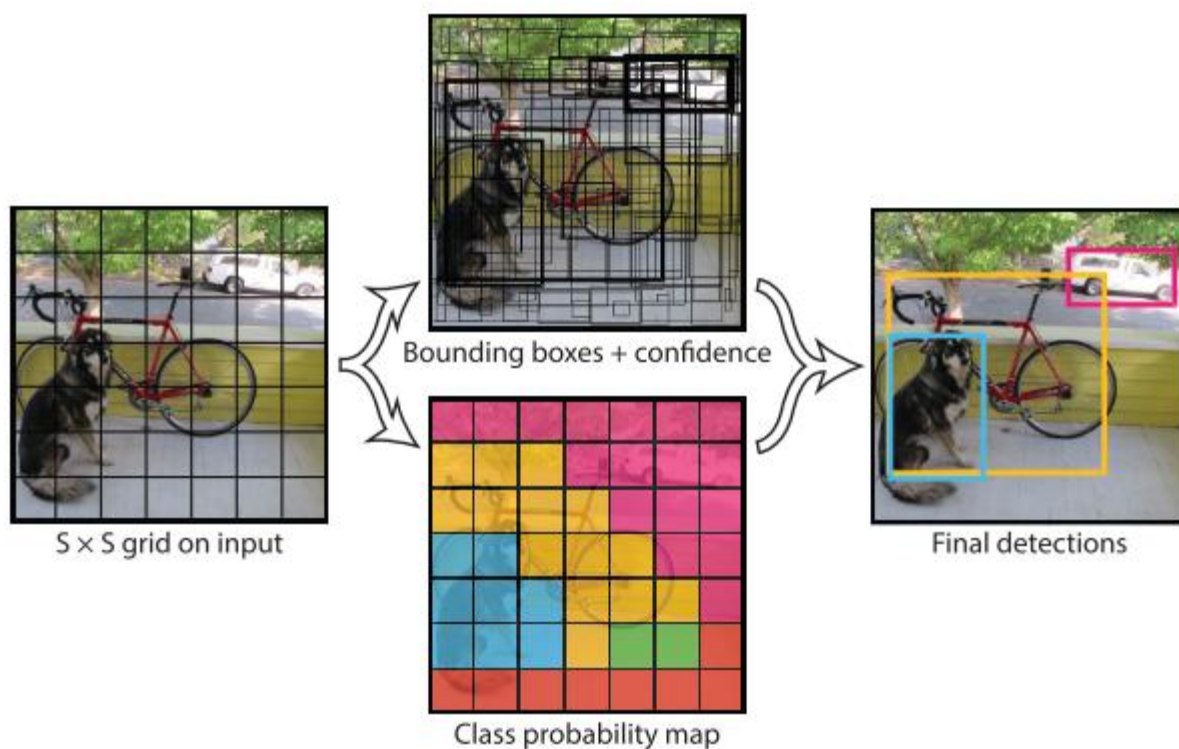
Redmon et al, "You Only Look Once: Unified, Real-Time Object Detection", CVPR 2016

Liu et al, "SSD: Single-Shot MultiBox Detector", ECCV 2016

Lin et al, "Focal Loss for Dense Object Detection", ICCV 2017

Object Detection

YOLO: Real-Time Object Detection



每个格子预设B个边界框

Within each grid cell:

- Regress from each of B base boxes with 5 numbers:
(dx, dy, dh, dw, **confidence**)
- Predict score for each of C classes

Output size:

$7 \times 7 (5 * B + C)$

Redmon et al, "You Only Look Once: Unified, Real-Time Object Detection", CVPR 2016

Liu et al, "SSD: Single-Shot MultiBox Detector", ECCV 2016

Lin et al, "Focal Loss for Dense Object Detection", ICCV 2017

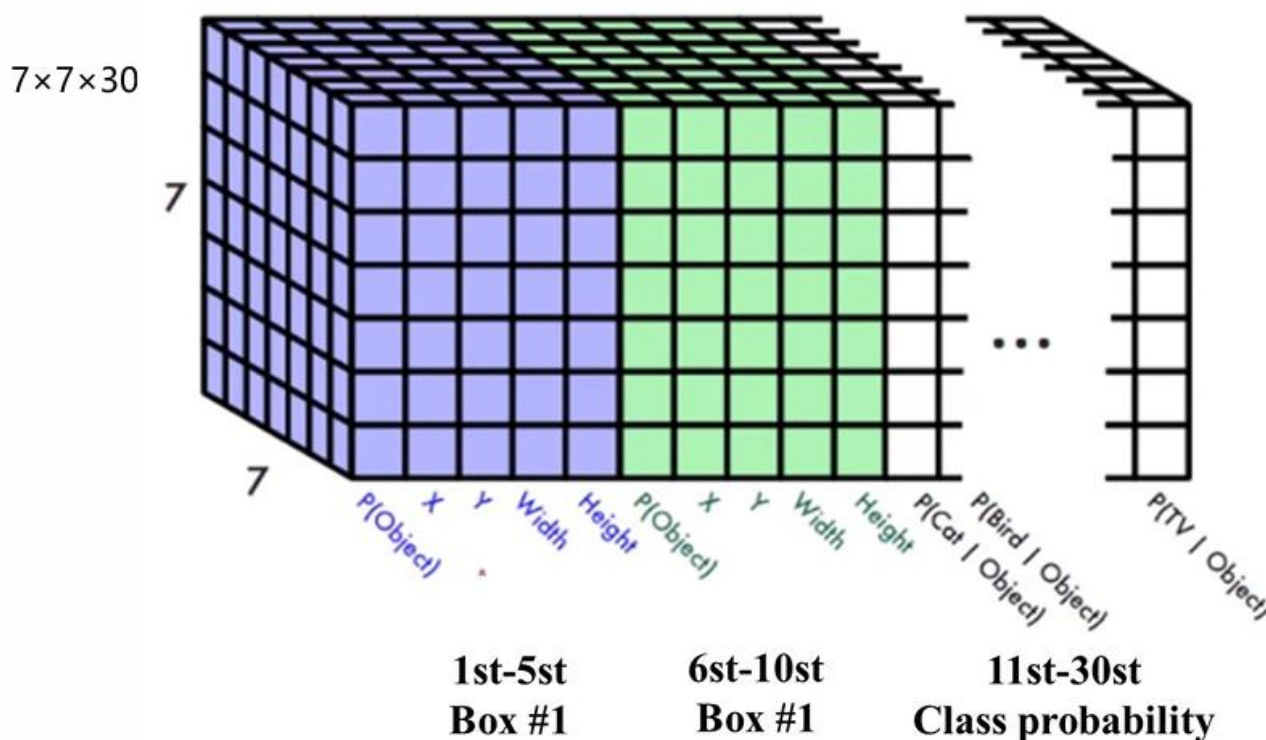
Object Detection

YOLO: Real-Time Object Detection

每个格子预设B个边界框

Within each grid cell:

- Regress from each of B base boxes with 5 numbers:
(dx, dy, dh, dw, **confidence**)
- Predict score for each of C classes



Output size:

$7 \times 7 (5 \cdot B + C)$

$B=2, C=20$

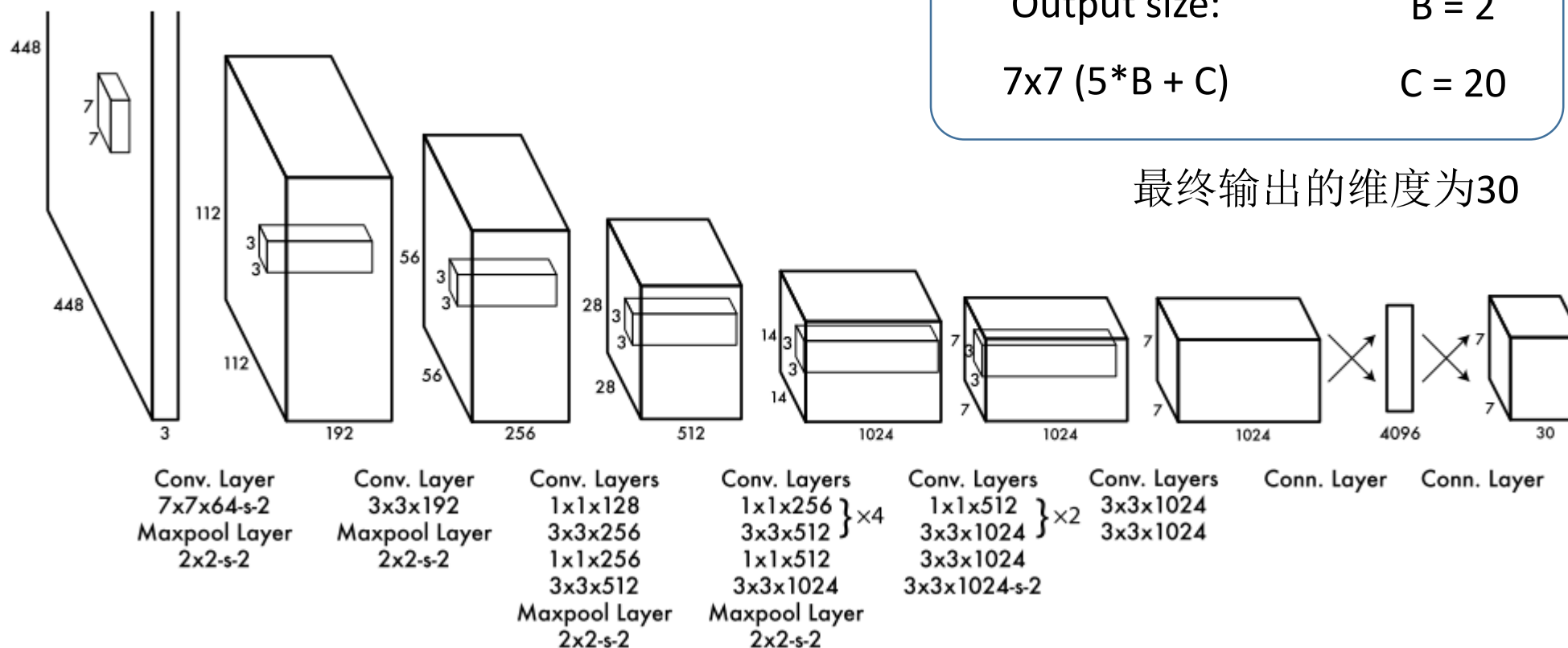
Redmon et al, "You Only Look Once: Unified, Real-Time Object Detection", CVPR 2016

Liu et al, "SSD: Single-Shot MultiBox Detector", ECCV 2016

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Object Detection

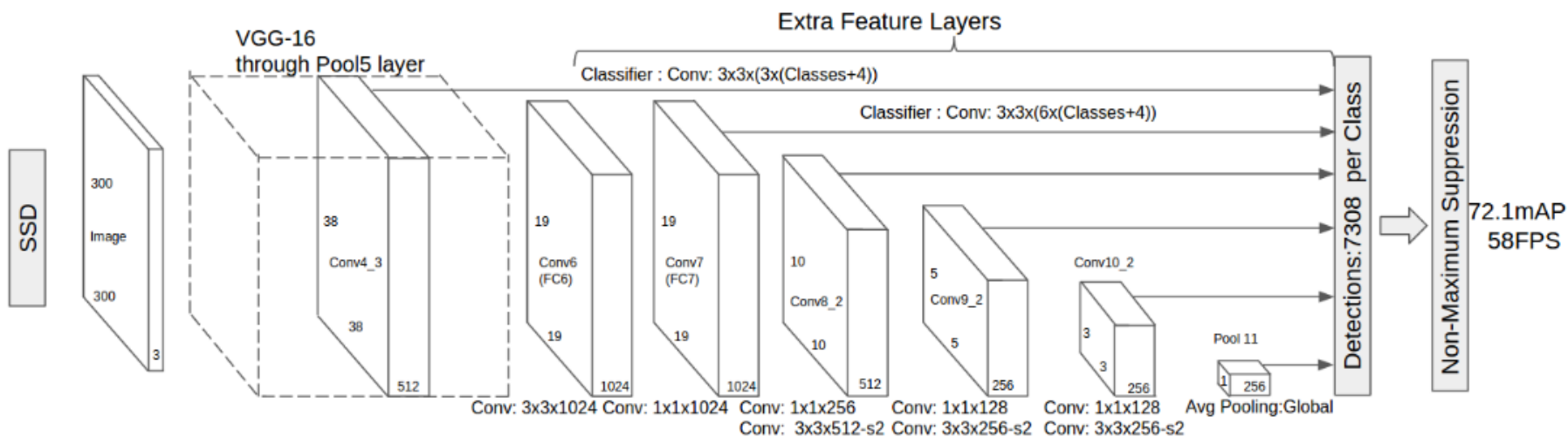
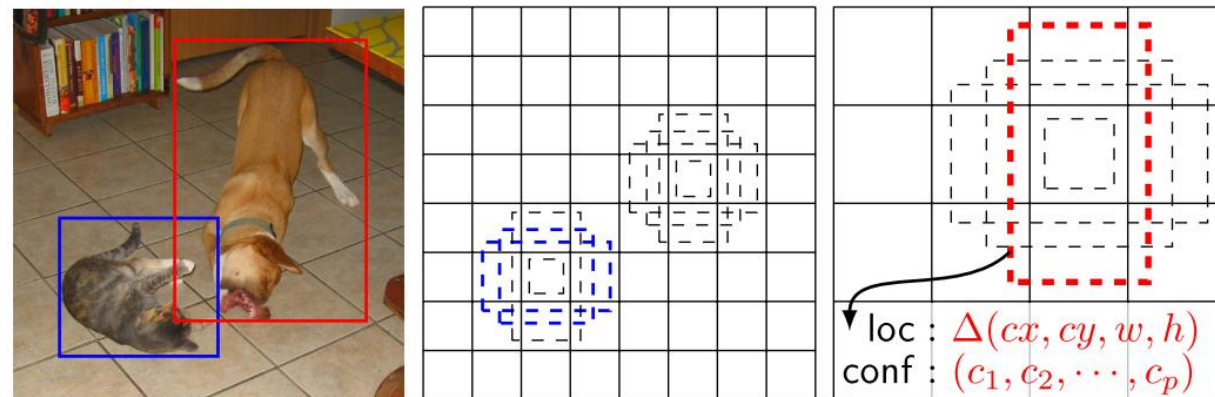
YOLO: Real-Time Object Detection



Object Detection

Single-stage detectors: SSD

- 多尺度预测
- 在预设边框（固定长宽比）的基础上进一步预测其与真实边框的偏移值



Object Detection

Single-stage detectors: SSD

- mAP可达到80%以上（精度高于Faster R-CNN）
- 并且可实时检测（效率高于YOLO）

Method	<i>mAP</i>	FPS	# Boxes
Faster R-CNN [2](VGG16)	73.2	7	300
Faster R-CNN [2](ZF)	62.1	17	300
YOLO [5]	63.4	45	98
Fast YOLO [5]	52.7	155	98
SSD300	72.1	58	7308
SSD500	75.1	23	20097

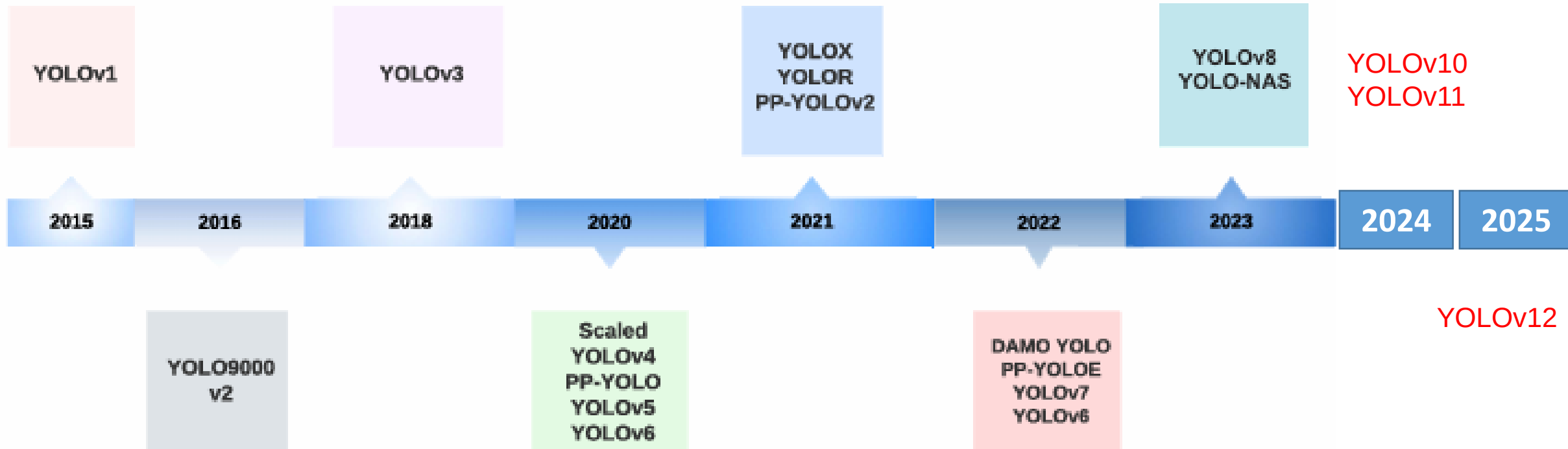
Results on Pascal VOC2007 test



Object Detection



Timeline of YOLO versions



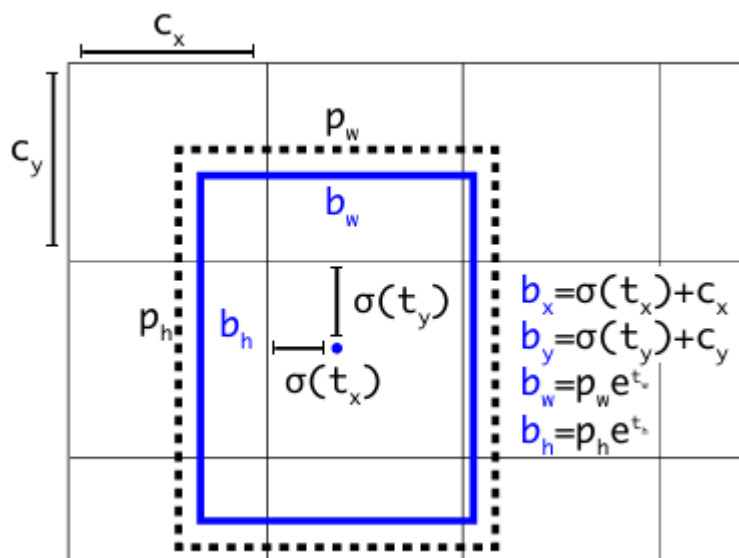
Object Detection



YOLOV-v3 (2018)

新增：多尺度预测、使用更深的网络

支持不同尺度的物体的检测、更强的通用性



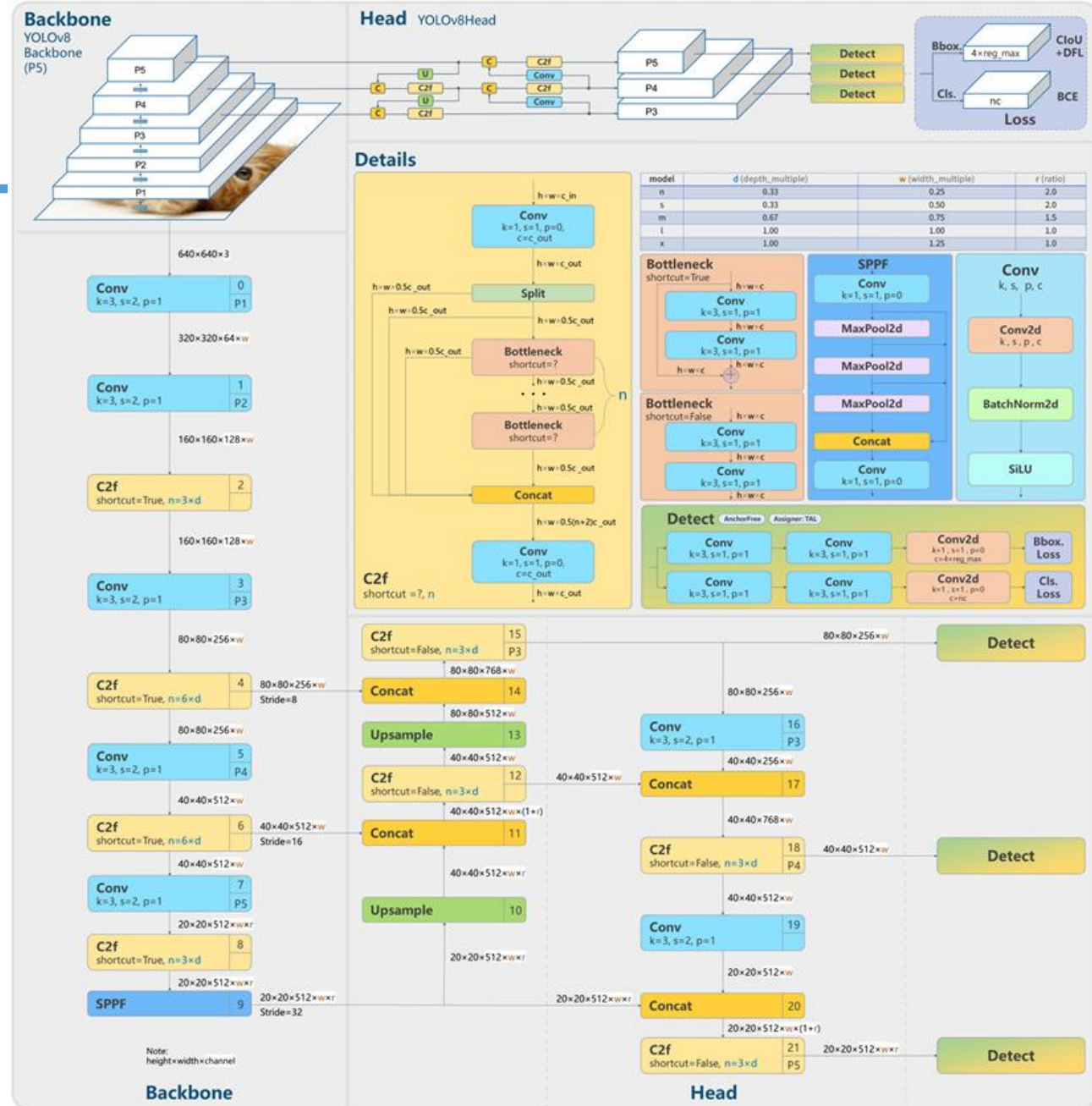
	Type	Filters	Size	Output
1x	Convolutional	32	3 × 3	256 × 256
	Convolutional	64	3 × 3 / 2	128 × 128
	Convolutional	32	1 × 1	
	Convolutional	64	3 × 3	
	Residual			128 × 128
2x	Convolutional	128	3 × 3 / 2	64 × 64
	Convolutional	64	1 × 1	
	Convolutional	128	3 × 3	
	Residual			64 × 64
	Convolutional	256	3 × 3 / 2	32 × 32
8x	Convolutional	128	1 × 1	
	Convolutional	256	3 × 3	
	Residual			32 × 32
	Convolutional	512	3 × 3 / 2	16 × 16
	Convolutional	256	1 × 1	
8x	Convolutional	512	3 × 3	
	Residual			16 × 16
	Convolutional	1024	3 × 3 / 2	8 × 8
	Convolutional	512	1 × 1	
	Convolutional	1024	3 × 3	
4x	Residual			8 × 8
	Avgpool		Global	
	Connected		1000	
	Softmax			

Darknet-53

Object Detection

YOLOV-v8 (2023)

- **全新的视觉模型**：保持精度的同时，实现了较高的检测速度，并且同时支持多种视觉任务。
- **新的骨干网络**：使用梯度流更丰富的C2f结构，并对不同尺度模型调整了不同的通道数，大幅提升了模型性能。
- **解耦头**：Head部分从原先的耦合头变成了解耦头结构，简化了模型结构并提高了推理速度。
- **损失函数**：采用了TaskAlignedAssigner正样本分配策略，并引入了Distribution Focal Loss确保了检测结果的准确性和鲁棒性。



小结

- Object detection
 - R-CNN, Fast R-CNN
 - YOLO and SSD

